

Obai Gasmalla

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Professional Summary

Data Consultant with 4+ years delivering data and analytics for **transport and infrastructure** clients and asset owners across rail, highways, the wider built environment and energy networks, underpinned by an MSc in **Artificial Intelligence** (Distinction). I engage stakeholders, design solutions, develop in **Python** and **SQL**, and present clearly to technical and non-technical audiences. Delivery for **National Highways**, **Network Rail** and **Transport for Wales** spans digital asset management, asset valuation and predictive modelling, using **statistical methods**, **Power BI** dashboards and data visualisation in **Tableau**, and production pipelines on **Azure** and **Databricks** with **Git**, **CI/CD** and **MLflow**.

Technical Skills

Programming Languages: **Python** (pandas, polars, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow), **SQL**, **R**, **C#**; OOP, TDD

Statistical Methods & Machine Learning: statistical analysis and data mining, regression, classification, clustering (elbow method, silhouette score), time-series forecasting, feature engineering, model evaluation, NLP (embeddings, transformers, TF-IDF), GBMs (XGBoost, LightGBM)

Data Visualisation & Reporting: **Power BI**, **Tableau**, Matplotlib, Seaborn, ggplot2; dashboard design, data storytelling

Location Intelligence & Geospatial: geospatial data analysis (**pandas**, **GeoPandas**), GIS data processing and mapping, reconciliation of location-based asset registers, transport data pipelines

Data Engineering & Cloud: ETL/ELT pipelines, **Databricks**, Apache Spark, Azure Data Factory, ML Studio and Functions, PostgreSQL, Microsoft Fabric, data quality and governance frameworks; **Microsoft Azure** (primary), AWS, GCP

Engineering & MLOps: **Git/GitHub**, **CI/CD**, automated testing (pytest, Behave), **MLflow**, Docker, model deployment and monitoring

Consulting & Delivery: client-facing delivery across the project lifecycle, stakeholder engagement, requirements gathering, commercial awareness, use-case prioritisation by business impact, **agile delivery (Scrum/Kanban)**, user training and adoption, mentoring, knowledge sharing

Professional Experience

Data Consultant (Data Scientist), WSP, London, United Kingdom Jul 2024 - Present

- Deliver client-facing analytics for infrastructure clients across the full delivery lifecycle: engage stakeholders to gather requirements and determine root causes, design solutions, develop and test in **Python** and **SQL**, and present outcomes tailored to technical and non-technical audiences
- **Design and implement** the ETL/ML pipeline for structures asset valuation on **Spark**, **Databricks** and **Azure**, deploying predictive models end to end with automated testing, **model evaluation and monitoring frameworks** and **CI/CD**
- Interrogate **millions of asset records** using **SQL** and **Python**, replacing a manual, laptop-dependent process with a ~20-minute automated workflow feeding downstream valuation and predictive-maintenance models
- Engineer **NLP pipelines** and statistical models over unstructured free-text data on structures (**transformers**, **embeddings**, TF-IDF), automating critical analysis workflows with **10/10 client feedback**
- Own data science initiatives end to end, translating business questions into analytical problems and **prioritising use cases by business impact**, aligning stakeholders and delivery teams on **measurable outcomes**
- On the **East West Rail Digital Spine Programme (MWJV)**, develop **ETL pipelines** integrating **DOORS** requirements data into the Digital Spine and own the **semantic layer architecture**, defining common business entities and data models for cross-system traceability across **Azure**, **PostgreSQL** and **Microsoft Fabric**
- Engineer an **Azure Function App** in **C#** (HTTP-triggered, managed identity), replacing a Logic App that could not handle large file transfers and hardening cross-system data movement
- Deliver **Power BI** dashboards, reporting and **user training** so non-technical stakeholders and executives monitor outputs self-serve
- Support the upgrade of the **CIRTAS** road collision data system to meet **STATS19** requirements for **Gateshead Council**, authoring test summary documentation and running **30+ tests** across data imports, exports and **GIS mapping** features
- **Coach and mentor junior team members** through code reviews and knowledge sharing, contributing to team growth

- Provide **input to tenders and bids**, supporting the identification and pursuit of new client opportunities, including bid and proposal activity that supported the extension of asset-valuation work

Graduate Data Consultant, WSP, London, United Kingdom

Oct 2022 - Jul 2024

- Developed **large-scale analytical pipelines** in **Python** and **R** that link defects to recommendations and fast-track engineering decisions on **Network Rail** and **National Highways** work
- Implemented **data quality** assurance and metadata management frameworks ensuring compliance with **data governance** and regulatory requirements
- Worked alongside modellers, engineers and clients in multidisciplinary delivery teams, presenting findings and supporting data-driven decisions; designed and implemented **transport data pipelines** for Transport for West Midlands (TfWM)
- Contributed as a **Data Engineer** to **Transport for Wales (TfWRL)** commercial analytics and service performance integration into **Azure**, delivering **Power BI** training so end users could extract and analyse commercial insights self-serve

Selected Projects

National Highways Asset Management System (AMS) Data Migration and Transformation, WSP

Jan 2026 - Present

- Deliver **data transformation and migration** workstreams, developing **Python**, **pandas** and **GeoPandas**-based solutions to reconcile asset registers and datasets across multiple source systems
- Design automated **data quality**, conflict-resolution and reporting workflows, including dependency analysis, change tracking and exception management, reducing manual effort and improving confidence in AMS-ready datasets

Energy Demand Forecasting, Northern Powergrid x Newcastle University, Turing Institute Data Study Group

2022

- Performed **exploratory analysis** on electricity distribution network data, identifying cost trends and **proposing a time-series forecasting model** for energy demand to support net-zero transition planning
- Delivered reproducible analysis in Jupyter Notebooks with **GitHub**, collaborating with the network's operational architects as part of the **Turing Institute Data Study Group**

Wind Turbine Surface Defects Detection, University of West London

2022

- MSc dissertation developing a **deep learning-based computer vision** algorithm for automated wind turbine surface defect detection, applying convolutional neural networks and transfer learning with rigorous performance evaluation

Education

Artificial Intelligence MSc (Computer Vision), Grade: Distinction (First Class)

University of West London,
London, United Kingdom

Relevant Coursework: Computer Vision, Deep Learning Architectures, Advanced Machine Learning, Statistical Methods, Data Mining, Natural Language Processing, Research Methodologies

Electrical Engineering BSc (Hons), specialisation in Control Theory

Sudan University of Science
and Technology, Sudan

Guest Lecturer (Lecture Support), University of West London (2022 - 2023): supported the module lead in delivering lectures and classes to undergraduate and postgraduate Computer Science students

Publication: Musa, M., Gasmalla, O., et al. (2024). *Distributed Geospatial Information Systems: Challenges and Opportunities*. In Exploring Remote Sensing: Methods and Applications

Professional Certifications

Microsoft Azure: AZ-900 Azure Fundamentals and AI-900 Azure AI Fundamentals

ITIL 4 Professional: IT service management and delivery

IBM Data Science Professional Certificate: Analyse Data with Python and Python Basics for Data Science

Languages

English (Fluent), Arabic (Fluent)